

Whiteboard

1 LASSO decomposition

Given a weight $\alpha_1 > 0$, consider the following Lasso model:

$$\omega_1 := \min_{\beta} \quad ||X\beta - y||_2^2 + \alpha_1 ||\beta||_1,$$

which we can recast as a conic program

$$\omega_1 = \min_{\beta, \eta, \eta_1} \quad \eta^2 + \alpha_1 \eta_1 \tag{1}$$

$$\text{s.t.} \quad ||X\beta - y||_2 \leq \eta \tag{2}$$

$$||\beta||_1 \leq \eta_1 \tag{3}$$

1.1 Relaxing the least squares

Pricing Relaxing (2) in the objective with a dual penalization $(\lambda, \lambda_0) \in \mathcal{L}_2^{n+1}$ the pricing becomes

$$\omega_1^{\text{ls}}(\lambda_0, \lambda) = \min_{\beta, \eta} \quad \eta^2 + \alpha_1 ||\beta||_1 - \lambda_0 \eta - \lambda^\top (y - X\beta)$$

The problem in η has a closed form solution $\eta = \lambda_0/2$, so that after some rearrangements we obtain:

$$\omega_1^{\text{ls}}(\lambda_0, \lambda) = \min_{\beta} \left\{ \alpha_1 ||\beta||_1 + (X^\top \lambda)^\top \beta \right\} - \lambda^\top y - \frac{\lambda_0^2}{4},$$

which is a big, but simple LP. Remark that when solving the next pricing problem, only the linear part of the objective changes and the last pricing's optimal basis is usable to reoptimize, probably requiring only a few simplex pivots.

Master The master problem with a set of K columns β^k becomes

$$\begin{aligned} \omega_1^{\text{ls}}(\mathcal{S}^k) = \min_{\theta} \quad & \left\| \sum_{k=1}^K \theta_k X \beta^k - y \right\|_2^2 + \alpha_1 \left\| \sum_{k=1}^K \theta_k \beta^k \right\|_1 \\ \text{s.t.} \quad & e^\top \theta = 1, \quad \theta \geq 0 \end{aligned}$$

where $\mathcal{S}^k := \text{conv}(\beta^k)_{k \in [K]}$.

Nota: Creo que no es necesario relajar nada más que la norma de los least squares. Es el único coupling constraint entre las distintas variables b_i . Si se relaja, el problema pasa a ser completamente independiente entre los b_i , que es el caso ideal (completamente paralelizable). Este es el gol general de los clásicos métodos de decomposición.

1.2 Relaxing the ℓ_1 cone

Pricing Relaxing (3) in the objective with a dual penalization $(\mu, \mu_0) \in \mathcal{L}_\infty^{n+1}$ the pricing becomes

$$\omega_1^{\ell_1}(\mu_0, \mu) = \min_{\beta, \eta_1} \{ \|X\beta - y\|_2^2 + \alpha_1 \eta_1 - \mu_0 \eta_1 - \mu^\top \beta \},$$

which, if the dual variables (μ_0, μ) are optimal for some master problem (meaning that $\alpha_1 = \mu_0$), can reduce to:

$$\omega_1^{\ell_1}(\mu_0, \mu) = \min_{\beta} \{ \|X\beta - y\|_2^2 - \mu^\top \beta \},$$

whose optimal solution has the following closed form expression: $\beta := (X^\top X)^{-1} (X^\top y + \frac{1}{2}\mu)$.

Master The master problem with a set of K columns β^k is exactly the same as before:

$$\begin{aligned} \omega_1^{\ell_1}(\mathcal{S}^k) = \min_{\theta} \quad & \left\| \sum_{k=1}^K \theta_k X \beta^k - y \right\|_2^2 + \alpha_1 \left\| \sum_{k=1}^K \theta_k \beta^k \right\|_1 \\ \text{s.t.} \quad & e^\top \theta = 1, \quad \theta \geq 0 \end{aligned}$$

1.3 Comments

- Hay que discutir si los η son variables que se descomponen tambien o no: ahi cambia harto el master para ambas relajaciones.
- El segundo pricing de ahi tiene valor acotado :-)

2 Boundedness of Pricing

Let us consider the following generic relaxation of the integer problem:

$$\min_{\beta, z, u, \xi} \quad \xi^2 + \tau e^\top z + \kappa e^\top u \tag{4}$$

$$\text{s.t.} \quad \|y - X\beta\|_2 \leq \xi \tag{5}$$

$$\left\| \begin{matrix} 2\beta_i \\ u_i - z_i \end{matrix} \right\|_2 \leq u_i + z_i, \quad \forall i \in [m] \tag{6}$$

$$u, z \geq 0 \tag{7}$$

2.1 Lagrangean dual

Associating dual variables $(\lambda_0, \lambda) \in \mathcal{L}_2^{n+1}$ to the constraint (5) and $(\pi_i, \gamma_i, \mu_i) \in \mathcal{L}_2^3$ to each constraint (6), the Lagrangean dual becomes

$$\max_{(\lambda, \lambda_0, \pi, \gamma, \mu) \in \mathcal{D}} \min_{z, u \geq 0, \beta, \xi} \left\{ \xi^2 + \tau e^\top z + \kappa e^\top u - \lambda_0 \xi - \lambda^\top (y - X\beta) - \sum_{i=1}^m (2\pi_i \beta_i + \gamma_i (u_i - z_i) + \mu_i (u_i + z_i)) \right\},$$

where $\mathcal{D} := \{(\lambda, \lambda_0, \pi, \gamma, \mu) : (\lambda_0, \lambda) \in \mathcal{L}_2^{n+1}, (\pi_i, \gamma_i, \mu_i) \in \mathcal{L}_2^3, \forall i \in [m]\}$. The dual is thus:

$$\begin{aligned}
& \max_{\lambda, \lambda_0, \pi, \gamma, \mu} && -\lambda_0^2/4 - \lambda^\top y \\
& \text{s.t.} && \|\lambda\|_2 \leq \lambda_0 \\
& && \left\| \begin{pmatrix} \pi_i \\ \gamma_i \end{pmatrix} \right\|_2 \leq \mu_i, && \forall i \in [m] \\
& && \tau + \gamma_i - \mu_i \geq 0, && \forall i \in [m] \\
& && \kappa - \gamma_i - \mu_i \geq 0, && \forall i \in [m] \\
& && (X^\top \lambda)_i = 2\pi_i, && \forall i \in [m],
\end{aligned}$$

which can be simplified as follows:

$$\begin{aligned}
& \max_{\lambda, \gamma, \mu} && -\|\lambda\|_2^2/4 - \lambda^\top y \\
& \text{s.t.} && \left\| \begin{pmatrix} (X^\top \lambda)_i \\ 2\gamma_i \end{pmatrix} \right\|_2 \leq 2\mu_i, && \forall i \in [m] \\
& && \tau + \gamma_i - \mu_i \geq 0, && \forall i \in [m] \\
& && \kappa - \gamma_i - \mu_i \geq 0, && \forall i \in [m] \\
& = \max_{\lambda, \gamma} && -\|\lambda\|_2^2/4 - \lambda^\top y \\
& \text{s.t.} && (X^\top \lambda)_i^2 \leq 4\tau(\tau + 2\gamma_i), && \forall i \in [m] \\
& && (X^\top \lambda)_i^2 \leq 4\kappa(\kappa - 2\gamma_i), && \forall i \in [m]
\end{aligned}$$

The bound on $(X^\top \lambda)_i$ is the loosest when both bounds are equal, i.e. when

$$\gamma_i = \frac{\kappa^2 - \tau^2}{2(\tau + \kappa)} = \frac{\tau - \kappa}{2},$$

giving:

$$\begin{aligned}
& \max_{\lambda} && -\|\lambda\|_2^2/4 - \lambda^\top y \\
& \text{s.t.} && (X^\top \lambda)_i^2 \leq 4\tau\kappa, && \forall i \in [m] \\
& = \max_{\lambda} && -\|\lambda\|_2^2/4 - \lambda^\top y \\
& \text{s.t.} && \|X^\top \lambda\|_\infty \leq 2\sqrt{\tau\kappa}
\end{aligned}$$

We can already see a pricing-boundedness criterion when fixing λ .

2.2 Equivalent Lasso model

In the last dual, we can also sketch an alternate proof for the lasso equivalence result. In fact, when dualizing it back in primal form, we obtain exactly a lasso model with parameter $2\sqrt{\tau\kappa}$.

2.3 Generating columns when the pricing is unbounded

When the pricing is unbounded, it is enough to provide an extreme ray of the pricing to the master: this can be done by normalizing it in some way, maybe some are better than others.

2.4 Why is the pricing bounded only at the final iteration?

[Todo renaud]

3 Elastic Net comparison

Given weights $\alpha_1, \alpha_2 > 0$, consider the following model:

$$\omega_{\text{net}} := \min_{\beta, z} \left\{ \|X\beta - y\|_2^2 + \alpha_1 \sum_{i=1}^m z_i + \alpha_2 \|\beta\|_2^2 : z \in \{0, 1\}^n, |\beta_i| \leq Mz_i, \forall i \in [m] \right\}$$

which we can recast as

$$\omega_{\text{net}} = \min_{\beta, z, u} \quad \|X\beta - y\|_2^2 + \alpha_1 \sum_{i=1}^m z_i + \alpha_2 \sum_{i=1}^m u_i \quad (8)$$

$$\text{s.t.} \quad |\beta_i| \leq Mz_i, \quad \forall i \in [m] \quad (9)$$

$$\beta_i^2 \leq u_i, \quad \forall i \in [m] \quad (10)$$

$$z_i \in \{0, 1\}, \quad \forall i \in [m] \quad (11)$$

3.1 Local Convexification

Consider the following set, corresponding to an individual set of constraints (9)-(10)-(11):

$$\mathcal{F} := \{(\beta, z, u) \in \mathbb{R} \times \{0, 1\} \times \mathbb{R} : |\beta| \leq Mz, \beta^2 \leq u\}$$

We now exhibit an explicit and compact form for $\text{conv}(\mathcal{F})$. A tuple (β, z, u) belongs to $\text{conv}(\mathcal{F})$ iff there is $\theta \in [0, 1]$ such that:

$$z = \theta \cdot 1 + (1 - \theta) \cdot 0$$

$$\beta = \theta\beta_1 + (1 - \theta)\beta_0$$

$$u = \theta u_1 + (1 - \theta)u_0$$

$$|\beta_1| \leq M \cdot 1, \quad |\beta_0| \leq M \cdot 0$$

$$\beta_1^2 \leq u_1, \quad \beta_0^2 \leq u_0$$

After some simplifications we obtain

$$z = \theta \in [0, 1]$$

$$\beta = z\beta_1$$

$$u = \theta u_1 + (1 - \theta)u_0$$

$$|\beta_1| \leq M, \quad \beta_0 = 0$$

$$\beta_1^2 \leq u_1, \quad 0 \leq u_0$$

or equivalently:

$$\text{conv}(\mathcal{F}) := \{(\beta, z, u) \in \mathbb{R} \times [0, 1] \times \mathbb{R} : |\beta| \leq Mz, \beta^2 \leq uz\}$$

which is SOC representable.

3.2 Elasticnet relaxation

Using the previous result, we obtain the following relaxation of our integer problem

$$\begin{aligned} \omega_{\text{net}} \geq \hat{\omega}_{\text{net}} &:= \min_{\beta, z, u} \quad ||X\beta - y||_2^2 + \alpha_1 \sum_{i=1}^m z_i + \alpha_2 \sum_{i=1}^m u_i \\ \text{s.t.} \quad &|\beta_i| \leq Mz_i, & \forall i \in [m] \\ &\beta_i^2 \leq u_i z_i, & \forall i \in [m] \\ &u_i \geq 0, \quad z_i \in [0, 1], & \forall i \in [m] \end{aligned}$$

where we can interpret each variable u_i as the magnitude of β_i^2 .

3.3 “Elasticnet” in “almost” lasso form

Using a similar duality trick as in the last section, we can prove that (keeping the $z_i \leq 1$ and bigM constraints), we have:

$$\begin{aligned} \tilde{\omega}_{\text{net}} &:= \min_{\beta, z, u} \quad ||X\beta - y||_2^2 + \alpha_1 \sum_{i=1}^m z_i + \alpha_2 \sum_{i=1}^m u_i \\ \text{s.t.} \quad &|\beta_i| \leq Mz_i, & \forall i \in [m] \\ &\beta_i^2 \leq u_i z_i, & \forall i \in [m] \\ &u_i \geq 0, \quad z_i \in [0, 1], & \forall i \in [m] \\ &= \max_{\pi, \lambda \geq 0} \left\{ \min_{\beta} \left\{ ||X\beta - y||_2^2 + \sum_{i=1}^n |\beta_i| \left(\pi_i + 2\sqrt{\alpha_2(\alpha_1 + \lambda_i - \pi_i M)} \right) \right\} - \lambda^\top e \right\} \end{aligned}$$

Comentarios:

- de hecho cada variable dual λ_i (resp. π_i) corresponde a la restriccion $z_i \leq 1$ (resp. $|\beta_i| \leq Mz_i$). Si no estuvieran esas restricciones, los λ, π desaparecen y nos llega el lasso equivalente.
- Mejor dicho: sin las cotas $z_i \leq 1$ y los bigM, si uno pesca el problema en (z_i, u_i) a β_i fijo, su valor optimo tiene forma cerrada.
- En otras palabras, si se guarda $z_i \leq 1$ o bigM: elasticnet $\neq$ lasso, pero se tiene la equivalencia si se relajan esas cotas sup. y bigM

3.4 Another formulation

An easy to implement Elastic Net (and its decomposition method) can be formulated adding the new term to the LASSO SOCP (4) formulation as follows

$$\min_{\beta, z, u, \xi_1, \xi_2} \quad \xi_1^2 + \tau e^\top z + \kappa e^\top u + \frac{1}{2}\theta \xi_2^2 \quad (12)$$

$$\text{s.t.} \quad \|y - X\beta\|_2 \leq \xi_1 \quad (13)$$

$$\left\| \begin{pmatrix} 2\beta_i \\ u_i - z_i \end{pmatrix} \right\|_2 \leq u_i + z_i, \quad \forall i \in [m] \quad (14)$$

$$\|\beta\|_2 \leq \xi_2 \quad (15)$$

$$u, z \geq 0, \quad (16)$$

with a master problem in some iteration k that can be written as

$$\min_{\pi, \xi_1, \xi_2} \quad \xi_1^2 + \tau \sum_{i=1}^m \mathbf{z}_i^k \pi + \kappa \sum_{i=1}^m \mathbf{u}_i^k \pi + \frac{1}{2}\theta \xi_2^2 \quad (17)$$

$$\text{s.t.} \quad \|y - X\mathbf{B}^k \pi\|_2 \leq \xi_1, \quad (18)$$

$$\left\| \begin{pmatrix} \mathbf{u}_i^k - \mathbf{z}_i^k \\ 2\mathbf{b}_i^k \end{pmatrix} \pi \right\|_2 \leq \mathbf{u}_i^k + \mathbf{z}_i^k, \quad \forall i \in [m] \quad (19)$$

$$\|\mathbf{B}^k \pi\|_2 \leq \xi_2, \quad (20)$$

$$\mathbf{Z}^k \pi, \mathbf{U}^k \pi \geq 0, \quad (21)$$

$$\pi \in \Pi. \quad (22)$$

The two pricing relaxations alternatives (maintaining the idea of the best LASSO relaxation) are:

$$\min_{\beta, z, u, \xi_1, \xi_2} \quad (\xi_1 - \mu_1^k)\xi_1 + (\psi^k{}^\top X - \phi^k)\beta + \tau e^\top z + \kappa e^\top u - \psi^k{}^\top y + \left(\frac{1}{2}\theta \xi_2 - \mu_2^k\right)\xi_2 \quad (23)$$

$$\text{s.t.} \quad \left\| \begin{pmatrix} 2\beta_i \\ u_i - z_i \end{pmatrix} \right\|_2 \leq u_i + z_i, \quad \forall i \in [m] \quad (24)$$

$$u, z \geq 0, \quad (25)$$

In this case, one can infer that the boundedness condition should be $\|\psi^k{}^\top X - \phi^k\|_\infty \leq 2\sqrt{\tau\kappa}$ (to do: check it).

$$\min_{\beta, z, u, \xi_1, \xi_2} \quad (\xi_1 - \mu_1^k)\xi_1 + \psi^k{}^\top X\beta + \tau e^\top z + \kappa e^\top u - \psi^k{}^\top y + \frac{1}{2}\theta \xi_2^2 \quad (26)$$

$$\text{s.t.} \quad \left\| \begin{pmatrix} 2\beta_i \\ u_i - z_i \end{pmatrix} \right\|_2 \leq u_i + z_i, \quad \forall i \in [m] \quad (27)$$

$$\|\beta\|_2 \leq \xi_2 \quad (28)$$

$$u, z \geq 0, \quad (29)$$

Since $\theta > 0$, this case should always attain an minimum, and it should be possible to obtain the solution without using a solver. Assuming the equality holds in the conic constraints on the optimal point, the solution for the main variable should be $\beta^* = -\frac{1}{\theta} \left(2\sqrt{\tau\kappa} + \psi^k{}^\top X \right)$.

4 Real data results

Implementation of the (12) model, it's decomposition with the pricing (23), and the comparison with the scikit-learn package.

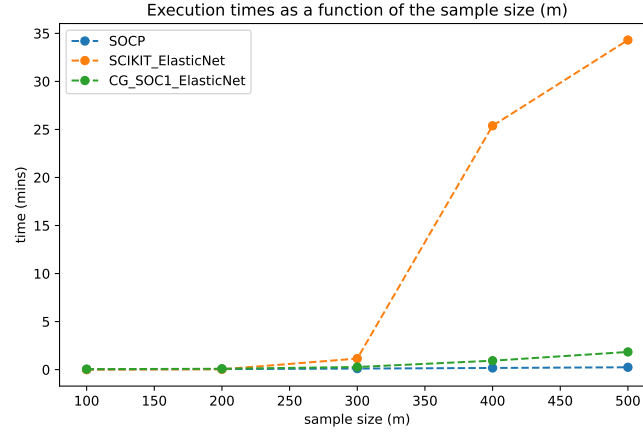


Figure 1: In the final two instances of *SCIKIT_ElasticNet*, coordinate-descent did not find the optimal solution in less than 10^6 iterations.

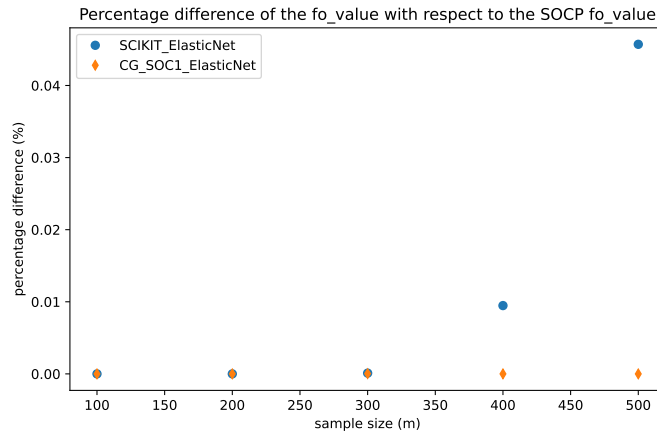


Figure 2: In the final two instances of *SCIKIT_ElasticNet*, coordinate-descent did not find the optimal solution in less than 10^6 iterations.

5 Numerical errors

Example of a clear problem with real data on the third iteration, **infeasible master problem**:

Numerical solver

```
(CVXPY) Jan 12 01:09:40 AM: Invoking solver MOSEK to obtain a solution.
```

```
(CVXPY) Jan 12 01:09:40 AM: Problem
(CVXPY) Jan 12 01:09:40 AM:   Name :
(CVXPY) Jan 12 01:09:40 AM:   Objective sense : maximize
(CVXPY) Jan 12 01:09:40 AM:   Type : CONIC (conic optimization problem)
(CVXPY) Jan 12 01:09:40 AM:   Constraints : 90
(CVXPY) Jan 12 01:09:40 AM:   Affine conic cons. : 0
(CVXPY) Jan 12 01:09:40 AM:   Disjunctive cons. : 0
(CVXPY) Jan 12 01:09:40 AM:   Cones : 2442
(CVXPY) Jan 12 01:09:40 AM:   Scalar variables : 18166
(CVXPY) Jan 12 01:09:40 AM:   Matrix variables : 0
(CVXPY) Jan 12 01:09:40 AM:   Integer variables : 0
(CVXPY) Jan 12 01:09:40 AM:
(CVXPY) Jan 12 01:09:40 AM: Optimizer started.
(CVXPY) Jan 12 01:09:40 AM: Presolve started.
(CVXPY) Jan 12 01:09:40 AM: Linear dependency checker started.
(CVXPY) Jan 12 01:09:40 AM: Linear dependency checker terminated.
(CVXPY) Jan 12 01:09:40 AM: Eliminator — tries : 0
time : 0.00
(CVXPY) Jan 12 01:09:40 AM: Lin. dep. — tries : 1
time : 0.00
(CVXPY) Jan 12 01:09:40 AM: Lin. dep. — number : 0
(CVXPY) Jan 12 01:09:40 AM: Presolve terminated. Time: 0.03
(CVXPY) Jan 12 01:09:40 AM: Optimizer terminated. Time: 0.06
(CVXPY) Jan 12 01:09:40 AM:
(CVXPY) Jan 12 01:09:40 AM:
(CVXPY) Jan 12 01:09:40 AM: Interior-point solution summary
(CVXPY) Jan 12 01:09:40 AM:   Problem status : PRIMAL INFEASIBLE
(CVXPY) Jan 12 01:09:40 AM:   Solution status : PRIMAL INFEASIBLE_CER
(CVXPY) Jan 12 01:09:40 AM:   Dual. obj: -4.7974632372e+06 nrm: 2e+00
Viol. con: 0e+00 var: 4e+00 cones: 0e+00
```

Summary

```
(CVXPY) Jan 12 01:09:40 AM: Problem status: unbounded
(CVXPY) Jan 12 01:09:40 AM: Optimal value: -inf
(CVXPY) Jan 12 01:09:40 AM: Compilation took 9.591e+00 seconds
(CVXPY) Jan 12 01:09:40 AM: Solver (including time spent in interface) took 1.475e-01
The optimal value is -inf
```

Doesn't solve the issue:

- Rounding decimals of: $X, y, (X^\top \beta), \beta$
- Changing the convergence tolerance

Does solve some issues:

- Solved some SOCP problems standardizing X and y . Still have some problems in CVXOPT with $m = 1500$, and real data:

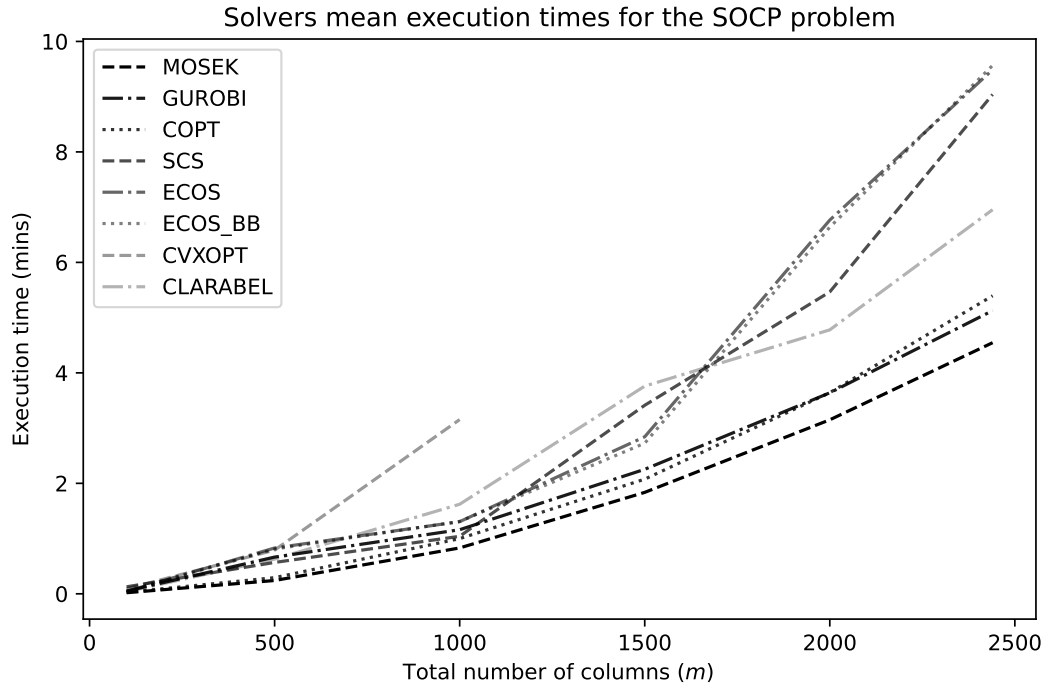
Numerical solver

(CVXPY) Jan 14 04:44:51 PM: Invoking solver CVXOPT to obtain a solution.

	pcost	dcost	gap	pres	dres	k/t
0:	-7.4270e-14	-1.0000e+00	6e+03	1e+00	1e+02	1e+00
1:	4.8808e-02	-1.2476e-02	2e+02	7e-02	5e+00	4e-02
2:	2.1662e-02	2.0155e-02	5e+00	2e-03	2e-01	1e-03
3:	2.1815e-02	2.1697e-02	4e-01	1e-04	1e-02	1e-04
4:	2.1778e-02	2.1775e-02	1e-02	5e-06	4e-04	3e-06
5:	2.1777e-02	2.1777e-02	2e-04	6e-08	4e-06	4e-08
6:	2.1777e-02	2.1777e-02	6e-06	2e-09	2e-07	5e-10

Terminated (singular KKT matrix).

Note: if you have an infeasible problem, you will either get the "singular KKT matrix" error, or the solver will terminate after the maximum number of iterations.



Problem with CG-LC1 and COPT in iteration 5:

Numerical solver

(CVXPY) Jan 15 10:55:28 PM: Invoking solver COPT to obtain a solution.
Cardinal Optimizer v7.0.5. Build date Jan 4 2024
Copyright Cardinal Operations 2023. All Rights Reserved

Setting parameter 'Logging' to 1
Setting parameter 'AbsGap' to 1e-06
Model fingerprint: e2a15a48

Using Cardinal Optimizer v7.0.5 on Windows
Hardware has 8 cores and 16 threads. Using instruction set X86_NATIVE (1)
Minimizing an SOCP problem

The original problem has:
4126 rows, 4138 columns and 744349 non-zero elements
102 cones
The presolved problem has:
3688 rows, 4107 columns and 743796 non-zero elements
102 cones

Starting barrier solver using 8 threads

Problem info:
Range of matrix coefficients: [6e-06,1e+00]
Range of rhs coefficients: [6e-05,2e+00]
Range of bound coefficients: [0e+00,0e+00]
Range of cost coefficients: [1e+00,1e+03]

Factor info:
Number of dense columns: 392
Number of matrix entries: 7.484e+05
Number of factor entries: 8.253e+05
Number of factor flops: 1.784e+08

Iter	Primal.Obj	Dual.Obj	Compl	Primal.Inf	Dual.Inf
Time					
0	+2.39159503e+04	-1.00000000e+00	6.38e+02	2.79e+00	1.17e+03
0.27s					
1	+2.34525698e+04	-1.12114546e+01	8.44e+01	2.74e+00	1.14e+03
0.30s					
2	+2.15486580e+04	-6.74099743e+01	1.70e+01	2.52e+00	1.06e+03
0.35s					
3	+1.08105031e+04	-3.03175393e+02	1.63e+00	1.31e+00	5.49e+02
0.38s					
4	+5.68904935e+03	-5.47263621e+01	6.27e-01	6.81e-01	2.85e+02

0.42 s						
5	+3.79365999e+03	+2.13624283e+02	3.68e−01	4.24e−01	1.77e+02	
0.46 s						
6	+1.79867938e+03	+5.96601840e+02	1.25e−01	1.42e−01	5.93e+01	
0.50 s						
7	+1.41006133e+03	+9.17993951e+02	5.68e−02	5.79e−02	2.42e+01	
0.53 s						
8	+1.26924311e+03	+1.02083020e+03	3.10e−02	2.92e−02	1.22e+01	
0.58 s						
9	+1.20030026e+03	+1.07695684e+03	1.62e−02	1.45e−02	6.07e+00	
0.64 s						
10	+1.15124182e+03	+1.09372214e+03	7.71e−03	6.77e−03	2.83e+00	
0.69 s						
11	+1.13188671e+03	+1.10638667e+03	3.51e−03	3.00e−03	1.25e+00	
0.74 s						
12	+1.11736147e+03	+1.11054791e+03	9.45e−04	8.01e−04	3.35e−01	
0.77 s						
13	+1.11392798e+03	+1.11229530e+03	2.28e−04	1.92e−04	8.03e−02	
0.82 s						
14	+1.11281650e+03	+1.11259605e+03	3.09e−05	2.59e−05	1.08e−02	
0.85 s						
15	+1.11274235e+03	+1.11261822e+03	1.74e−05	1.46e−05	6.05e−02	
0.94 s						
16	+1.11265673e+03	+1.11264376e+03	1.82e−06	1.52e−06	1.80e−02	
1.01 s						
17	+1.11264687e+03	+1.11264670e+03	2.96e−08	2.02e−08	2.66e+01	
1.08 s						

Barrier status:	NUMERICAL
Primal objective:	1.11281650e+03
Dual objective:	1.11259605e+03
Duality gap (abs/rel):	2.20e−01 / 1.98e−04
Primal infeasibility (abs/rel):	2.59e−05 / 2.24e−05
Dual infeasibility (abs/rel):	1.08e−02 / 9.30e−06

Postsolving

Solving finished
Status: Numerical Objective: — Iterations: 17 Time: 1.10 s